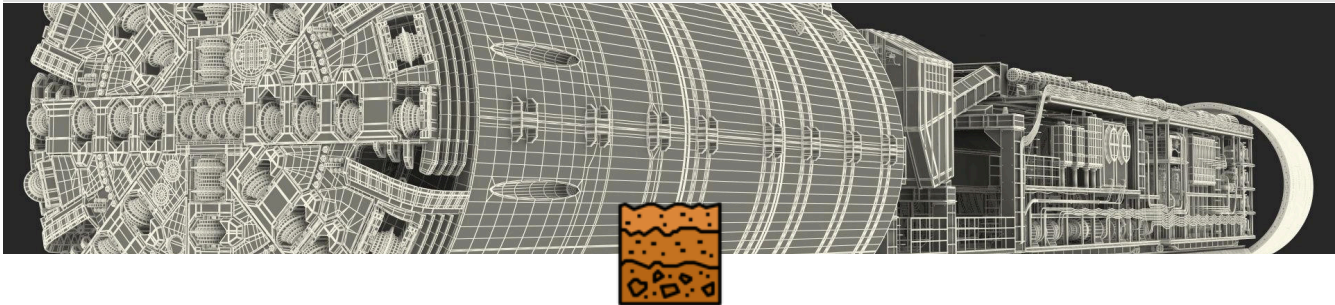




Soil Classification - Singapore Cr111

Description

This project implements a deep learning model using Convolutional Neural Networks (CNNs) in TensorFlow to automatically classify different soil types from image data.

The system addresses key challenges such as inconsistent lighting, object interference, and dataset redundancy through pre-processing, camera setup optimisation, and data cleaning techniques. The model is trained on labelled soil images and demonstrates the potential for use in geotechnical site assessment for TBM. Related research validates the proposed architecture and supports the methodology applied.

We are currently collaborating with the on-site team in Singapore to collect relevant soil image data, which includes Old Alluvium types. In the meantime, the model will be trained using publicly available datasets sourced from platforms such as Kaggle.

Project Tracker

Current Progress (Updated Daily)

Date	Progress	Sources/PI C
11 Jun 2025	Briefed by @liew kit shen and @Farah Adibah Binti Badardin on new task to develop LLM soil detection	@Lilith Lee



	Worked on project research and documentation.	
12 Jun 2025	Completed research report and presented to Mr Sam and Ms Farah.  Project Research and Plan <u>Suggested improvements are sent to site team through email:</u> 1. <i>Install a top-down camera with zoom to capture soil detail.</i> 2. <i>Reduce image-capture frequency to reduce redundancy, capture only during constant soil flow.</i> 3. <i>Install flash or lighting for consistent image quality.</i>	@Lilith Lee
13 Jun 2025	Downloaded dataset from Kaggle, combined dataset from different sources. Cleaned up dataset and is currently dividing it into test and validation sets. (70% - 15% - 15% ratio) Meeting with on-site team to discuss potential camera setups and fund allocation.  13.6.25 Meeting Informed of potential collaboration with Singapore team.	@Lilith Lee
16 Jun 2025	Finished dividing up datasets locally. Finished setting up confluence for internal communication with Singapore team. Meeting with Ms. @CHONG CHEA GEE regarding current progress.  16.6.25 Meeting Setup python with libraries. Finished file-fetching logic.	@Lilith Lee
17 Jun 2025	Debugged TensorFlow virtual environment issue (version incompatibility). Normalised dataset by changing all images into jpg, while decreasing size. Debugged and verified file-fetching logic with image-counting capabilities.	@Lilith Lee

2025

available datasets and site-obtained datasets with Mr. Sam and Ms. Farah.

Completed model creation using keras packages. Test run indicates ~70% accuracy.

Included image batch inspection for data validation.

Background reading to develop ability to explain AI model verbally to non-technical stakeholders for senior-level operations.

comprehensive introduction:

[An Introduction to Convolutional Neural Networks: A Comprehensive Guide to CNNs in Deep Learning](#)

Lee

19 Jun
2025

read up on potential data extraction methods to convey soil types distribution and decided on t-SNE method. T-SNE is chosen over PCA because it is more sensitive towards outliers, which will be useful in data analysis. This exploratory analysis can be used to find outliers in the dataset or identify poor class separation.

[Difference between PCA VS t-SNE - GeeksforGeeks](#)

Finished t-SNE extraction logic.

Explored potential of integrating heatmap capabilities to visualise deep learning by model.

@Lilith
Lee

20 Jun
2025

Debugged t-SNE extraction logic.

Researched machine learning heatmapping to gain a better understanding of CNN models.

[Get Heatmap from CNN \(Convolution Neural Network \), AKA CAM](#)

<https://ieeexplore.ieee.org/document/9374393>

[Fig. 3 Example heat map explanations for a CNN with image inputs. The...](#)

2025

Read up on t-SNE visualisation and fine-tuning.

Complete introduction: [How to Use t-SNE Effectively](#)

[Understanding t-SNE by Implementation | Towards Data Science](#)

[t-SNE clearly explained](#)

Currently fine-tuning graph for better visualisation.

[Data Visualisation Tracker](#)

During discussion with Mr. Sam, identified current need to discuss with the site team about the potential use cases for the model we are developing.

Informed of need to develop soil identification LLM model for comparison purposes.

Lee

24 Jun
2025

Completed t-SNE visualisation fine-tuning. Graphs are in [Data Visualisation Tracker](#).

Set up GCS ADC for file-transfer logic. Completed GCS to local logic.

Working on TBM image pre-processing logic.

@Lilith
Lee

25 Jun
2025

Completed pre-processing logic. Includes cropping, duplication removal and gaussian blur to remove noise (if required)

Logic is coded in modular fashion. Currently testing different parts of logic. Cropping and gaussian blur is working as intended.

Tested GCS to local logic, working as intended.

Details of file transfer:

Period: 26th May- 25th June (30 days)

Total files: 2831 files

only WB files are downloaded.

@Lilith
Lee

2025	Was briefed by Ms. Farah on Slurry Simulation Project after this project. Was informed that the file download mechanism should include sort function based on ring number. Then files are labelled respective soil types with Ms. Chea Gee's help.	Lee
30 Jun 2025	Completed debugging for file duplication logic. Updated file-downloading mechanism to include ring number based folder-organisation. Downloaded files from GCS and uploaded to personal GCD, ready for labelling with Ms. Chea Gee's help.  GCS Dataset	@Lilith Lee
1 Jul 2025	Shared drive link with Ms. Chea Gee and team. Was informed that soil type can be labelled using PFTA. (Geological profile is requested by Mr. Wei Meng to be updated in the intervention report)	@Lilith Lee
3 Jul 2025	Was informed by Ms Farah that PFTA graph is not ready yet. Built and debugged LLM logic as an alternative method to CNN framework. LLM framework utilises Gemini 2.5 flash to generate a prompt.	@Lilith Lee
4 Jul 2025	Updated prompt with context and clearer instructions. The prompt now includes a numerical evaluation of confidence and types of soil to help narrow down identification. Numerical output will be exported as excel and plotted using python. <u>output example:</u> file: 01060602previewen-02b.jpg <i>name='1' colour='Reddish-brown to brownish-orange' wetness='Dry' size='Poorly graded, ranging from sand to gravel and cobbles' confidence='10'</i> -----	@Lilith Lee

	<i>name='1' colour='Light brown to yellowish-tan' wetness='Dry' size='Coarse-grained, a mix of sand, gravel, and cobbles' confidence='10'</i> ----- file: 1.jpg <i>name='9' colour='Reddish-brown' wetness='Dry' size='Aggregated/Clumpy' confidence='9'</i> ----- file: 10.jpg <i>name='2' colour='Light yellowish-brown' wetness='Very dry' size='Fine to medium sand' confidence='8'</i> Was informed by Mr Sam of another project with higher urgency. Will put this project on hold until the other project is completed.	
7 Jul 2025	project still on hold.	
8 Jul 2025	other project is completed. Was briefed by Ms. Farah about PFTE graphs and labelling techniques.	
10 Jul 2025	Labelled and cleaned the datasets for training. Downloaded all available images and sorted according to Ring Number and WB/EB. Dataset images are cropped and duplicates are removed. Attached is an example of the cleaned dataset. 	@Lilith Lee

	PDF graphs for other ring numbers are obtained, we can label the other images to get a larger dataset.	
11 Jul 2025	python --version: Python 3.10.11 ran CNN model with cleaned dataset. ~83% accuracy achieved. Model is ready for export and it's application is to be determined. Git repository in Bitbucket is established.	@Lilith Lee
14 Jul 2025	Source code updated in Bitbucket. Code is now available for pull.	@Lilith Lee
15 Jul 2025	documentation for main_CNN.py is established in confluence.	@Lilith Lee
16 Jul 2025	documentation for all python scripts are completed. Please refer to Project Documentation for more details.	@Lilith Lee
	updated morning of next-day	

Recently Updated Content

-  [Soil Classification - Singapore Cr11](#)
Aug 13, 2025 • contributed by [Lilith Lee](#)
-  [Project Documentation](#)
Aug 12, 2025 • contributed by [Lilith Lee](#)
-  [gcs_download_specific_and_cleanin g.py Documentation](#)
Jul 16, 2025 • contributed by [Lilith Lee](#)
-  [gcs_download_all.py Documentation](#)
Jul 16, 2025 • contributed by [Lilith Lee](#)
-  [merge_folder.py Documentation](#)
Jul 16, 2025 • contributed by [Lilith Lee](#)
-  [format_converter.py Documentation](#)
Jul 16, 2025 • contributed by [Lilith Lee](#)
-  [main_LLM.py Documentation](#)
Jul 16, 2025 • contributed by [Lilith Lee](#)
-  [main_CNN.py Documentation](#)
Jul 15, 2025 • contributed by [Lilith Lee](#)
-  [Data Visualisation Tracker](#)
Jun 26, 2025 • contributed by [Lilith Lee](#)

Contributors

Lilith Lee



